

Lecture-25

Mathematics 1 (15B11MA111)

CO [C105.5]

Module: Ordinary Differential Equations

Topic: Homogeneous constant coefficient ODEs

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Differential Equations

[According to Wikipedia](#)

In mathematics, a differential equation is an equation that relates one or more functions and their derivatives. The study of differential equations is a wide field in pure and applied mathematics, physics, and engineering. All of these disciplines are concerned with the properties of differential equations of various types. Differential equations play an important role in modeling virtually every physical, technical, or biological process, from celestial motion, to bridge design, to interactions between neurons.

Two types of differential equations

Ordinary Differential Equations (ODE)

An ordinary differential equation (ODE) is a differential equation containing one or more functions of one independent variable and the derivatives of those functions.

$$\begin{array}{lll} 1. \frac{dy}{dx} = \frac{1+x^2}{1-y^2} & 2. \frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} - 8y = 0 & 3. \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}} = k \frac{d^2 y}{dx^2} \end{array}$$

Partial Differential Equations (PDE)

A partial differential equation (PDE) is a differential equation that contains unknown multivariable functions and their partial derivatives. (This is in contrast to ordinary differential equations, which deal with functions of a single variable and their derivatives.)

$$4. x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = mu, \quad 5. \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial z}{\partial y}$$

Order and Degree of a Differential Equation

The *order* of a differential equation is the order of the highest differential co-efficient present in the equation. Consider

$$1. \quad L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin \omega t.$$

$$2. \quad \cos x \frac{d^2 y}{dx^2} + \sin x \left(\frac{dy}{dx} \right)^2 + 8y = \tan x$$

$$3. \quad \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left(\frac{d^2 y}{dx^2} \right)^2$$

The order of the above equations is 2.

The degree of a differential equation is the degree of the highest derivative after removing the radical sign and fraction.

The *degree* of the equation (1) and (2) is 1. The degree of the equation (3) is 2.

Linear Differential Equation

A differential equation having the following properties is called as Linear Differential Equation:

- the dependent function y and its derivatives occur in the equation in the first degree only
- no products of y and/or any of its derivatives are present
- no transcendental functions – (trigonometric or logarithmic etc) of y or any of its derivatives occur

Otherwise it is called as non- linear differential equation.

Solution of a Differential Equation

An equation containing dependent variable (y) and independent variable (x) and free from derivative, which satisfies the differential equation, is called the solution (primitive) of the differential equation. The solution which contains arbitrary constants is called the general solution of the differential equation. The solution free from arbitrary constants i.e., the solution obtained from the general solution by giving particular values to the arbitrary constants is called a particular solution of the differential equation.

□ You have already studied methods for first order differential equations such as variable separable, Linear ODE etc. Now, we will discuss higher order ODEs (Second and higher order).

Higher Order Linear Differential Equations with Constant Coefficients

A linear ordinary differential equation of order n , is written as

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y = r(x)$$

or

$$a_0(x)y^{(n)}(x) + a_1(x)y^{(n-1)}(x) + \cdots + a_{n-1}(x)y'(x) + a_n(x)y(x) = r(x) \quad (1)$$

Where y is the dependent variable and x is the independent variable and $a_0(x) \neq 0$. If $r(x) = 0$, then it is called a *homogeneous equation*, otherwise it is called *non-homogeneous equation*.

For example, a second order homogeneous equation is of the form

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \quad a_0(x) \neq 0 \quad (2)$$

and a non-homogeneous second order equation is of the form

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = r(x), \quad a_0(x) \neq 0 \quad (3)$$

If $a_i(x), i = 0, 1, 2$ are constants then the equations are linear second order constant coefficient equations.

Differential Operator D

Sometimes, it is convenient to write the given linear differential equation in a simple form using the differential operator $D = d/dx$. We write

$$Df(x) = Df = (Df)(x) = \frac{df}{dx} = f'. \quad (3)$$

We have, for example $D(x^n) = \frac{d}{dx}(x^n) = nx^{n-1}$, n constant;

$D(\cos x) = -\sin x$, etc. Also,

$$D^3(f) = D(D^2 f) = D(f'') = f''', \dots, D^k(f) = f^{(k)}.$$

We define $D^0 \equiv 1$, so that if 1 is the operator defined by $1(f) = f$, we have $D^0(f) = 1(f) = f$.

We now define the operator L by

$$\begin{aligned}
 L &= a_0(x) \frac{d^n}{dx^n} + a_1(x) \frac{d^{n-1}}{dx^{n-1}} + \cdots + a_{n-1}(x) \frac{d}{dx} + a_n(x) \\
 &= a_0(x)D^n + a_1(x)D^{n-1} + \cdots + a_{n-1}(x)D + a_n(x) = P(D)
 \end{aligned} \tag{4}$$

which is a polynomial in D , so that

$$\begin{aligned}
 Ly &= a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y \\
 &= a_0(x)D^n y + a_1(x)D^{n-1}y + \cdots + a_{n-1}(x)Dy + a_n(x)y \\
 &= [a_0(x)D^n + a_1(x)D^{n-1} + \cdots + a_{n-1}(x)D + a_n(x)]y = P(D)y
 \end{aligned} \tag{5}$$

For example, the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$$

can be written as

$$Ly = (D^2 + 5D + 6)y = 0 \tag{6}$$

where the operator L is given by $L = P(D) = D^2 + 5D + 6$.

Solution of Higher Order Homogeneous Linear Equations with Constant Coefficients

Consider the n th order homogeneous linear equation with constant coefficients

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0 \quad (7)$$

We attempt to find a solution of the form $y = e^{mx}$.

Substituting $y = e^{mx}$, $y^{(k)} = m^k e^{mx}$, $k = 1, 2, \dots, n$ in Eq. (7)

and cancelling e^{mx} , we obtain the characteristic equation as

$$a_0 m^n + a_1 m^{n-1} + \cdots + a_{n-1} m + a_n = 0 \quad (8)$$

The degree of this algebraic equation is same as the order of the differential equation. This equation has n roots may be real and distinct, all or some of the roots may be equal, all or some of the roots may be complex.

Real and distinct roots

Let the polynomial (8) have all real and distinct roots $m_1, m_2, \dots, m_n$. Then the general solution is given by

$$y(x) = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x} \quad (9)$$

Example 1. Find the general solution of the differential equation

$$y''' - 2y'' - 5y' + 6y = 0.$$

Solution. Substituting $y = e^{mx}$, we obtain the characteristic equation as

$$m^3 - 2m^2 - 5m + 6 = 0.$$

The roots of this equation are $m = 1, -2, 3$. since the roots are real and distinct, the general solution of the equation is given by

$$y(x) = Ae^x + Be^{-2x} + Ce^{3x}.$$

Real multiple roots

The characteristic equation (8) may have some multiple roots. Let r be the multiplicity of the root m_1 . That is root $m = m_1$ is repeated r times. Let the remaining $n-r$ roots be real and distinct. Substituting $m = m_1$ we obtain $y_1(x) = e^{m_1 x}$ as one of the solutions. The remaining $r-1$ linearly independent solutions corresponding to the multiple root $m = m_1$ are given by

$$xy_1, x^2y_1, \dots, x^{r-1}y_1.$$

That is, the linearly independent solutions in this case are

$$e^{m_1 x}, xe^{m_1 x}, x^2e^{m_1 x}, \dots, x^{r-1}e^{m_1 x} \quad (10)$$

Example 2. Solve the differential equation $y''' - 3y' - 2y = 0$.

Solution. Substituting $y = e^{mx}$, we obtain the characteristic equation as

$$m^3 - 3m - 2 = 0, \text{ or } (m + 1)(m^2 - m - 2) = 0$$

or $(m + 1)^2(m - 2) = 0$, or $m = -1, -1, 2$.

Corresponding to the double root $m = -1$, the linearly independent solutions e^{-x} and xe^{-x} . Hence, the general solution is

$$y(x) = Ae^{2x} + (Bx + C)e^{-x}.$$

Complex roots

The complex roots occur in conjugate pairs and are of the form $p \pm iq$. Then, the solution of the equation can be written as

$$\begin{aligned}y(x) &= Ae^{(p+iq)x} + Be^{(p-iq)x} \\&= Ae^{px}e^{iqx} + Be^{px}e^{-iqx} \\&= (Ae^{iqx} + Be^{-iqx})e^{px} \\&= [A(\cos qx + i\sin qx) + B(\cos qx - i\sin qx)]e^{px}\end{aligned}$$

Simplifying, we obtain

$$y(x) = e^{px}[c_1 \cos qx + c_2 \sin qx]$$

where $c_1 = A + B$ and $c_2 = i(A - B)$.

• **Example 3.** Find the solution of the initial value problem

$$y'' + 4y' + 13y = 0, \quad y(0) = 0, y'(0) = 1.$$

Solution. The characteristic equation is given by

$$m^2 + 4m + 13 = 0,$$

$$\text{or } m = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i = p + iq.$$

The general solution is given by

$$y(x) = [A \cos qx + B \sin qx]e^{px} = e^{-2x}[A \cos 3x + B \sin 3x]$$

Substituting in the initial conditions, we obtain

$$y(0) = 0 = A,$$

$$y'(x) = Be^{-2x}(3 \cos 2x - 2 \sin 3x),$$

$$y'(0) = 1 = 3B, \text{ or } B = 1/3.$$

The solution of the initial value problem is

$$y(x) = (e^{-2x} \sin 3x)/3.$$

• Multiple complex roots

This case is a combination of the two earlier cases of real multiple roots and simple complex roots. Now, if $p + iq$ is a multiple root of order m , then $p - iq$ is also a multiple root of order m . For example, if $p_1 + iq_1$ is a double root, then $p_1 - iq_1$ is also a double root. The corresponding linearly independent solutions are

$$e^{p_1 x} \cos q_1 x, e^{p_1 x} \sin q_1 x, x e^{p_1 x} \cos q_1 x, x e^{p_1 x} \sin q_1 x.$$

Example 4. Solve the differential equation $y^{iv} + 32y'' + 256y = 0$.

Solution. The characteristic equation is

$$m^4 + 32m^2 + 256 = 0, \text{ or } (m^2 + 16)^2 = 0.$$

The roots of this equation are the double roots $m = \pm 4i$.

Therefore, the general solution is

$$y(x) = (Ax + B) \cos 4x + (Cx + D) \sin 4x.$$

Note: The solution of homogeneous part is called as Complementary Function (C. F.) of the differential equation.

Thank You